

Database Assignment 01
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Question 11:-

As LybiaNova is using file based system it may face several problems which includes:-

(i) Data-Manipulation/Duplication:-

Each branch stores its own customer & service records which leads to:-

- (a) Same customer stored multiple times.
- (b) Wasted storage space.

(ii) Data-Inconsistency:- What if one branch updates customer information & others do not which eventually results in:-
inaccuracy.

(iii) Slow data Access:-

To retrieve the whole information of a client, data needs to be collected from different branches which wastes time.

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(i) Peer Data sharing:-

Branches can't easily share the information.

(v) Weak Security & Control:-

file systems have limited access control.

→ changes made by someone can't be tracked.

(vi) No proper Backup & recovery:-

Data may be lost permanently if a file is corrupted or deleted.



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Question #1 (part b):

By using DBMS, many problems can be solved which results in:

(i) Centralized Data Storage:

All database/branches would use one central database. & it means no duplication & same updation for everyone.

(ii) Reduced Data Redundancy:

Data stored once & referenced when needed which saves space & improves accuracy.

(iii) Consistency: If one branch updates a record. All branches see the updated version instantly.

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(v) faster Data Retrieval:-

Using:-
SQL queries
indexing
Optimized searching.

(iv) Improved Security:-

DBMS provides:-
authentication
Role based access control
Data encryption.

(vi) Backup & Recovery:-

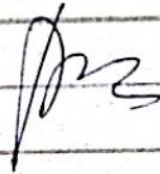
DBMS supports:-
Automatic Backup
Data-recovery mechanism.

(vii) Data-integrity Enforcement:-

Constraints like:-

- primary-key.
- foreign-key
- Not Null

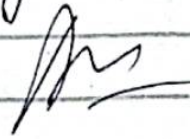
Ensure valid & reliable data.



Question #1 (Part c):-

A Centralized server based mobile application would:-

- (i) provide real-time access.
- (ii) Enable remote Accessibility.
- (iii) Synchronize all branches.
- (iv) Improves decision-making.
- (v) Secure Access control



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Question #2:

1. Select * from clients
Where country = 'Pakistan';
2. Select name, service-cost
from services
where service-cost > 500;
3. Select * from Service-Records
where status = 'Completed';
4. Select name/client.name
from clients
Where country = 'Pakistan' OR country =
'United States';
5. Select * from services
where service-cost BETWEEN 300 AND
1000;

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6. Select * DISTINCT location
from offices;

7. Select * from Service-Records
order by service-date desc;

8. Select Upper(client-name) as client-upper
from clients;

9. Select Count(*) as total-service-records
from service-records;

10. Select service-name, service-cost
order by service-cost asc

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Question #3..

1. Insert into Borrow-Record (Member-ID, Book-ID) values (104, TO-DATE(2026-02-05, "YYYY-MM-DD"), 1);

2. Update Book
Set Copies-Available = 1
Where Book-ID = 2;

3. Delete from Members
Where Member-ID = 101;

4. Insert INTO Members (Member-Name, email, Membership-Type) values (NULL, 'faizan@email.com', 'Premium');

5. Update Borrow-Record
Set Book-ID = 5
Where Record-ID = 1003;

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6. Insert into Books (Book-ID, Title, Author, Copies-Available values (4, 'Network-Security', 2);

7. Update Borrow-Records
Set Member-ID, 102
Where Record-ID, 1003;

8. Delete from Books
where Book-ID, 3;

9. Insert into Borrow-Records (Record-ID, Member-ID, Book-ID, Borrow-Date, Return-Date) values
(1001, 102, 3, '2026-02-07', '2026-02-15');

10. Alter table Books add constraint
check-copies check (Copies-Available > 51);

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Question #4 :-

1. There are 7 (seven) entities:-

1. Book
2. BOOK-AUTHORS
3. PUBLISHER
4. BOOK-COPIES
5. BOOK-LOANS
6. LIBRARY-BRANCH
7. BORROWER.

2. Primary Keys:-

(i) Book :-

↳ Book-ID

(ii) Book-Author :-

Book-ID, Author Name

(iii) Publisher :-

↳ Name

(iv) Book-copies :-

Book-ID, Branch ID

(v) Book loans :-

Book ID, Branch ID, Card No

(vi) Library-branch :-

Branch ID

(vii) Borrower :-

↳ Card No

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3. foreign key & their purpose:

Book:-

↳ PublisherName

↳ PUBLISHER(Name)

→ connects book to publisher

Book.copies:-

↳ BookId

↳ BranchId

↳ library-branch (BranchId)

Book Author:-

BookId

→ links author to books

Book-loans:-

↳ BookId

↳ BranchId

↳ library-branch

(BranchId)

↳ cardNo

4. Domain of each attribute:-

Book:-

↳ BookId (INT)

↳ Title (Varchar(100))

↳ PublisherName (Varchar(100))

Publisher:-

↳ Name (Varchar(100))

↳ Address (Varchar(100))

↳ phone (Varchar(15))

Book Author:-

↳ BookId (INT)

↳ AuthorName (Varchar(100))

Book.copies:-

↳ BookId (INT)

↳ BranchId (INT)

↳ No. of copies (INT)

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Book Loan :-

- ↳ Due Date (DATE)
- ↳ BookId (INT)
- ↳ BranchId (INT)
- ↳ CardNo (INT)
- ↳ DateOut (DATE)

Library branch :-

- ↳ BranchId (INT)
- ↳ BranchName (varchar)
- ↳ Address (varchar)

Borrower :-

- ↳ CardNo (INT)
- ↳ Name (varchar(100))
- ↳ Address (varchar(100))
- ↳ phone (varchar(100))

5. Relationship Type :-

relationship as :-

- one book can have multiple authors.
- Each author entry belongs to one BookId.

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Question #15:-

1. Data-Independence:-

Data-independence means that changes in the structure of the database don't affect the application programs that use the data.

Types:-

- (i) Physical Data-independence.
- (ii) Logical Data-independence.

2. Real-life Example:-

Ex small grocery store:-

→ File-system is better here:-

- (i) Very small amount of data.
- (ii) Single person access.
- (iii) No complex relationships.
- (iv) Easy & low cost.

→ Using DBMS here could be:-

- (i) Expensive
- (ii) Requires installation & maintenance.
- (iii) Unnecessary for small data.

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3. Difference b/w Delete, Drop, Truncate:

(i) Delete:

- removes selected data
- can use where clause
- Don't remove Table-structure

(ii) Drop: Deletes the entire table.
Removes table-structure.

(iii) Truncate: removes all rows
can't use where clause
faster than delete.
Don't remove Table-structure.

↳ Referential-Integrity ensures that foreign key values must match a primary-key in another table.

→ Referential-Integrity keeps the database accurate & consistent.

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Q. Can a foreign key have null?

Ans.

Yes, it can have null.

It can be null if:-

- When relationship is optional
- The column is not defined as Null.

c) if foreign key value is Not Null, it must match a valid primary key.

d) if it is Null, it simply means no reference.